PART-1 : PHYSICS

SECTION-I

1) A radioactive element of mass number 208 at rest disintegrates by emitting an α-particle. If E is the energy of the emitted α-particle, the energy of disintegration is :-

(A) $\frac{52}{51}E$

(B) $\frac{51}{52}E$

(C) 52 E

(D) E

2) A metallic plate of surface charge density $+2\epsilon_0$ (in SI units) is irradiated with photons of wavelength 620 nm. The work function of metal is 1.5 eV. What maximum distance from the metal plate can the initially ejected photoelectrons go before turning back? (Assume photoelectrons to be emitted perpendicular to the plate)

(A) No photoelectrons will be emitted

(B) Photoelectrons will not turn back at all

(C) $\frac{1}{4}m$

(D) $\frac{1}{2}m$

3) **Statement-1 :** A sphere, a cube and a thin circular plate made of same material and of same mass are initially heated to 200°C, the plate will cool at fastest rate.

Statement-2 : Rate of cooling $\frac{\Delta T}{dt} = \frac{e\sigma A(T^4 - T_0^4)}{mC}$. Surface area is maximum for circular plate.

(A) Statement–1 is True, Statement–2 is True; Statement–2 is a correct explanation for Statement–1

(B) Statement–1 is True, Statement–2 is True; Statement–2 is not a correct explanation for Statement–1

(C) Statement–1 is True, Statement–2 is False.

(D) Statement–1 is False, Statement–2 is True

4) A laser gun of power 3 mW and mass 50 gm emits photons of wavelength 500 nm. It is in gravity free space and emits for one hour. Find the distance covered by gun due to recoil in this one hour (approx.) :-

(A) 1.3 mm

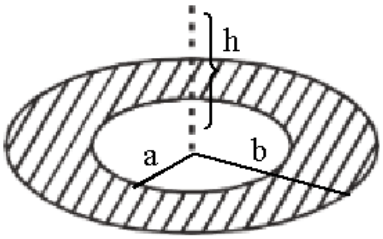
(B) 2.1 μm

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(C) 1.1 cm

(D) 8.5 mm

5) A thin disc of radius b = 2a has a concentric hole of radius 'a' in it (see figure). It carries uniform surface charge 'σ' on it. If the electric field on its axis at height 'h' (h << a) from its centre is given as 'Ch' then value of 'C' is :-



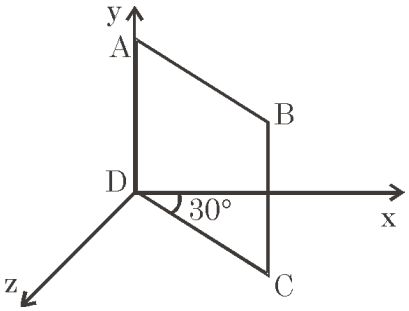
(A) $\frac{\sigma}{4a\varepsilon_0}$

(B) $\frac{\sigma}{2a\varepsilon_0}$

(C) $\frac{\sigma}{8a\varepsilon_0}$

(D) $\frac{\sigma}{a\varepsilon_0}$

6) A square cross-section ABCD of side a is kept in the direction shown in the diagram. An electric field $\vec{E} = \left(E_0\hat{i} + E_0\hat{j}\right)$ exists in the region. The flux of this field through the area of cross-section is:



(A) E_0a^2

(B) $\frac{E_0a^2}{3}$

(C) $\frac{E_0a^2}{2}$

(D) $\frac{\sqrt{3}}{2}E_0a^2$

7) Consider a solid insulating sphere of radius R with charge density varying as $\rho = \rho_0 r^2$ (ρ_0 is a constant and r is measure from centre). Consider two points A and B at distance x and y respectively (x < R, y > R) from the centre. If magnitudes of electric fields at points A and B are equal, then :-

(A) $x^2y = R^3$

(B) $x^3y^2 = R^5$

(C) $x^2y^3 = R^5$

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(D) $\frac{x^4}{y} = R^5$

8) A ring of radius R is made out of a thin wire of area of cross section A. The ring has a uniform charge Q distributed on it. A charge q_0 is placed at the centre of the ring. If Y is the young's modulus for the material of the ring and ΔR is the change in the radius of the ring then,

(A) $\Delta R = \frac{q_0 Q}{4\pi\varepsilon_0 RAY}$

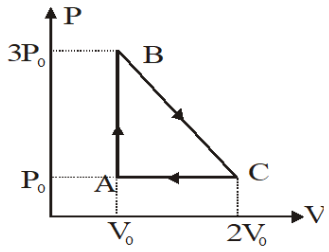
(B) $\Delta R = \frac{q_0 Q}{4\pi^2\varepsilon_0 RAY}$

(C) $\Delta R = \frac{q_0 Q}{8\pi^2\varepsilon_0^2 RAY}$

(D) $\Delta R = \frac{q_0 Q}{8\pi^2\varepsilon_0 RAY}$

9)

One mole of an ideal monoatomic gas is taken round the cyclic process ABCA as shown in figure.



List-I		List-II	
(P)	The work done by the gas during complete cycle	(1)	$\frac{25P_0V_0}{8R}$
(Q)	Heat absorbed by the gas in the path AB	(2)	$\frac{P_0V_0}{R}$
(R)	The minimum temperature attained by the gas during the cycle.	(3)	P_0V_0
(S)	The maximum temperature attained by the gas during the cycle.	(4)	$3P_0V_0$

(A) (P) - (4) ; (Q) - (2) ; (R) - (1) ; (S) - (3)

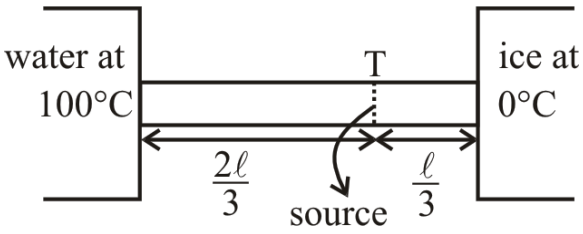
(B) (P) - (3) ; (Q) - (4) ; (R) - (2) ; (S) - (1)

(C) (P) - (1) ; (Q) - (4) ; (R) - (2) ; (S) - (3)

(D) (P) - (3) ; (Q) - (2) ; (R) - (1) ; (S) - (4)

10) The rod connecting two reservoir and connected to a source is as shown in diagram. The rod is in steady state. Find the temperature of source, so that rate of melting of ice is 16 times that rate of vaporization. (Latent heat of vaporization is 540 Cal/gm and latent heat of fusion is 80 Cal/gm)

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(A) 540° C

(B) 580° C

(C) 640° C

(D) 740° C

11) Determine the change in entropy ΔS of 100 gm of water when it is heated to the boiling point from room temperature of T = 300 K, and it is transformed into vapor at the boiling temperature and atmospheric pressure. It is given that for water : the molar heat capacity C = 75.4 J/(mol-K); latent heat of vaporization L = 2.26 MJ / kg.
[Take ln(1.24) = 0.217]

(A) 697 J/K

(B) 600 J/K

(C) 500 J/K

(D) 650 J/K

12) One mole of a diatomic gas undergoes a process $P = \frac{P_0}{1 + \left(\frac{V}{V_0}\right)^3}$ where P_0, V_0 are constants. The translational kinetic energy of the gas when $V = V_0$ is given by

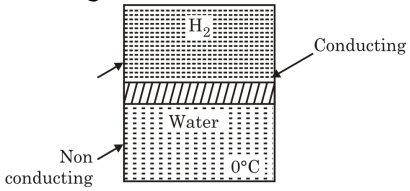
(A) $\frac{5P_0V_0}{4}$

(B) $\frac{3P_0V_0}{4}$

(C) $\frac{3P_0V_0}{2}$

(D) $\frac{5P_0V_0}{2}$

13) Initially all the water was at 0°C and the system having conducting separator separates water and H_2 gas as shown was left for sufficient time to attain thermal equilibrium. Finally only 10gm ice was found in the water chamber. If latent heat of fusion of ice is 80 cal/gm then translational K.E. of H_2 gas (Neglect thermal capacities of separator and container) :-



(A) Decreases by 480 cal

(B) Decreases by 800 cal

(C) Increases by 960 cal

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(D) Increases by 480 cal

14) Parallel rays of light of intensity I = 912 Wm^{-2} are incident on a spherical black body kept in surroundings of temperature 300 K. Take Stefan-Boltzmann constant $\sigma = 5.7\times10^{-8}$ Wm^{-2} K^{-4} and assume that the energy exchange with the surrounding is only through radiation. The final steady state temperature of the black body is close to–

(A) 330 K

(B) 660 K

(C) 990 K

(D) 1550 K

15) **Statement-1 :** Suppose that a Gaussian surface encloses no net charge. Gauss' law requires that electric field equals zero for all points on the surface.
Statement-2 : If electric field equals zero everywhere on the surface, Gauss' law requires that there be no net charge inside.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

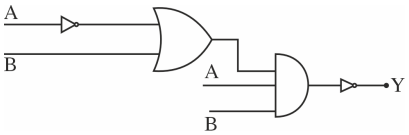
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

16)

Circuit given below is that of



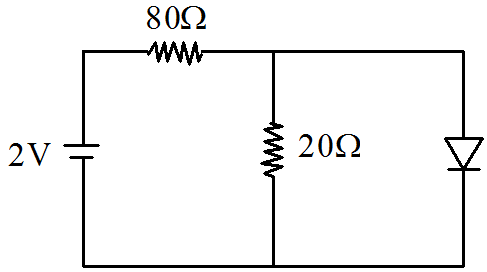
(A) AND gate

(B) OR gate

(C) NAND gate

(D) NOR gate

17) In the circuit shown, the current through the ideal diode is :-



(A) 75 mA

(B) 20 mA

(C) 100 mA

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(D) 25 mA

18) Two satellites are launched simultaneously into orbits of radius R and 4R. At an instant the two satellites are on the same radial line on the same side of earth. The time after which they have maximum separation is (mass of earth = M_e) Assume same sense of rotation.

(A) $\frac{8\pi}{7}\sqrt{\frac{R^3}{GM_e}}$

(B) $7\pi\sqrt{\frac{R^3}{GM_e}}$

(C) $\frac{7\pi}{8}\sqrt{\frac{R^3}{GM_e}}$

(D) $\pi\sqrt{\frac{R^3}{GM_e}}$

19) The little prince (the main character of the novel written by antoine de saint-Exupery) lives on the spherical planet named B-612, the density of which is 5200 kg/m^3. The Little Prince noticed that if he quickens his pace, he feels himself lighter. When he reached the speed of 2 m/s he became weightless, and began to orbit about the planet as a satellite. What is escape speed on the surface of planet ?

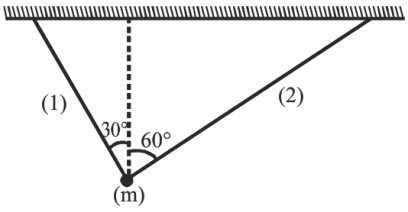
(A) $2\sqrt{2}$m/s

(B) 2 m/s

(C) $4\sqrt{2}$m/s

(D) $8\sqrt{2}$m/s

20) A point mass m is suspended using two wires of different material as shown in the figure. If cross-section area of wire-1 and wire-2 are 3 mm^2 and $\sqrt{3}$ mm^2 respectively, which of the following is correct?



(A) stress in wire -1 > stress in wire-2

(B) stress in wire -1 < stress in wire-2

(C) stress in wire -1 = stress in wire -2

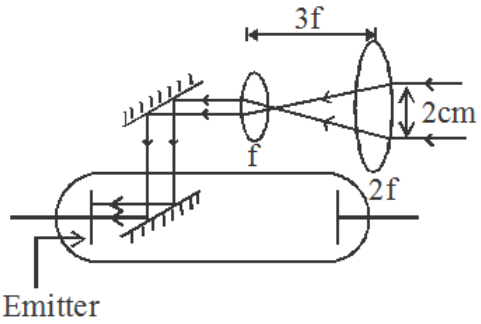
(D) value of Young's modulus of both the wires is needed

SECTION-II

1) Consider a hydrogen-like ionised atom with atomic number Z with a single electron. In the emission spectrum of this atom, the photon emitted in the n = 2 to n = 1 transition has energy 74.8 eV higher than the photon emitted in the n = 3 to n = 2 transition. The ionisation energy of the hydrogen atom is 13.6 eV. The value of Z is ..... .

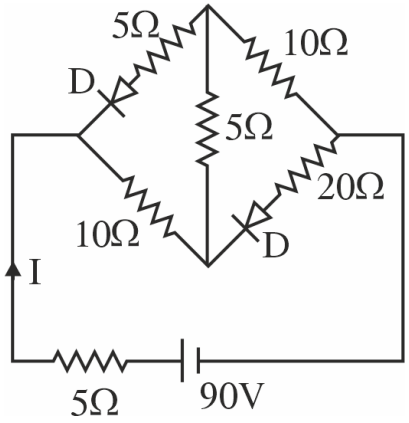
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2) A 2 cm thick parallel beam of light of frequency 1.6×10^{15} Hz and power 6.62 watt is directed on emitter as shown in figure. Neglect any loss of energy during reflection and refractions. If work function of emitter is 6 eV then the number of photons incidenting on emitter plate per second is K × 10^{18} , find 100K. ($h = 6.62 \times 10^{-34}$ JS)



3) In a refrigerator, heat from inside at 250 K is transferred to a room at 300 K. How many joules of heat shall be delivered to the room for each joule of electrical energy consumed ideally?

4) The current I (in A) in the network is:



5) A spherical ball of radius 7m is made up of a material having linear expansion coefficient $\alpha = 2 \times 10^{-5}$/°C and bulk modulus 10^4 Pa. If ball is heated by 20°C then calculate the needed pressure (Pa) to keep its volume constant.

PART-2 : CHEMISTRY

SECTION-I

1) A currency counting machine counts 60 million notes per day. A bank has as many notes as number of oxygen atoms in 24.8 g of $Na_2S_2O_3.5H_2O$ (Molar mass = 248 g mol^{-1}, $N_A = 6 \times 10^{23}$). How many days would be required to count these notes ?

(A) 9×10^{17}

(B) 7×10^{10}

(C) 8.0×10^{15}

(D) 6.6×10^{12}

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2) Select the number of correct statement regarding freezing point of following aqueous solution.
I. 0.1m $CaCl_2$ II. 0.1m $K_3[Fe(CN)_6]$ III. 0.1m NaCl IV. 0.1m $AlCl_3$
(Assume that all electrolyte are completely ionised in their solution)
Statement-1 : $CaCl_2$ solution is freezes at higher temperature than $K_3[Fe(CN)_6]$ solution.
Statement-2 : NaCl solution is freezes at higher temperature than $K_3[Fe(CN)_6]$ solution.
Statement-3 : Both $AlCl_3$ and $K_3[Fe(CN)_6]$ solution freezes at same temperature.
Statement-4 : $CaCl_2$ solution is freezes at lower temperature than $AlCl_3$ solution.

(A) 1
(B) 2
(C) 3
(D) 4

3) For 1 mole of an ideal monoatomic gas on moving from one state to another state, the temperature of gas is doubled and its pressure become 4 times. Then entropy change (ΔS) in the process will be :

(A) $\frac{R}{2}\ln 2$
(B) $-\frac{R}{2}\ln 2$
(C) $2R \ln 2$
(D) $\frac{5R}{2}\ln 2$

4) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$, $\Delta H < 0$
Select the correct statement for above reaction.

(A) At the start of the reaction, formation of $SO_3(g)$ takes place spontaneously.
(B) Decrease in total pressure will result in formation of more moles of $SO_3(g)$.
(C) K_p of reaction increases with formation of more mole of $SO_3(g)$ by increasing pressure.
(D) Reaction is entropy driven.

5) 2.5 g of a non-volatiles, non-electrolyte solute is dissolved in 100 g water at 25°C. The solution show boiling point 102°C. The vapour pressure of the solution at 100°C is :
Given : Molal boiling point elevation constant of water = 0.5 K.kg/mole, $P_{external}$ = 1 atm = 760 mmHg

(A) 720 mmHg
(B) 736 mmHg
(C) 698 mmHg
(D) 709 mmHg

6) **Statement-1 :** pOH of pure water decreases on increasing temperature.
Statement-2 : Dissociation of pure water is endothermic.

(A) Statement-1 is true, Statement-2 is true and Statement-2 is correct explanation for Statement-1
(B) Statement-1 is true, Statement-2 is true and Statement-2 is NOT the correct explanation for Statement-1

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(C) Statement-1 is true, Statement-2 is false

(D) Statement-1 is false, Statement-2 is true

7) Select the correct set of species with paramagnetic nature

(A) NO , O_2^+ , O_2^- , B_2

(B) B_2 , O_2 , N_2 , N_2^+

(C) O_2 , O_2^+ , O_2^{-2} , N_2^+

(D) Be_2 , He_2 , C_2 , N_2

8) Match the list:

List- I (Molecular species)		List- II (Properties)	
(P)	ClO_4^-	(1)	Planar and polar
(Q)	NH_2^-	(2)	All bond lengths are identical
(R)	SOF_4	(3)	Axial d-orbitals participates in hybridisation
(S)	$XeOF_4$	(4)	Non planar and non polar
		(5)	p_π-d_π bond(s) are present

The correct option is

(A) P-3; Q-2; R-1; S-5

(B) P-4; Q-2; R-5; S-3

(C) P-2; Q-5; R-2; S-1

(D) P-5; Q-4; R-2; S-1

9) Among the given statements.

(I) Octet completion is main driving force for $ICl_3 \rightarrow I_2Cl_6$

(II) If $B_2H_6 + 2L \rightarrow 2[H_3B \leftarrow L]$ then L can be $R\text{–}NH_2$

(III) Molecules $Al_2(CH_3)_6$ have 3c–2e$^-$ bond.

(IV) In $\dot{C}F_3$ unpaired electrons exist in sp^3 hybrid orbital.

How many of the above statements are correct?

(A) 2

(B) 3

(C) 1

(D) 4

10) Correct statement/s about hydrides of 16th group elements

(A) Reducing property increases H_2S To H_2Te

(B) Boiling point order is $H_2O > H_2Te > H_2Se > H_2S$

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(C) Only H_2O is a liquid at room temperature

(D) All are correct

11) Tin reacts with dilute nitric acid and produces:

(A) NO_2

(B) NO

(C) NH_4NO_3

(D) N_2

12) Which of thc following is INCORRECT about borax ?

(A) Produce alkaline solution in water

(B) Produce buffer solution in water

(C) Produce blue bead on heating with $CaSO_4$

(D) Used as an acidic flux

13) Extra pure N_2 can be obtained by heating

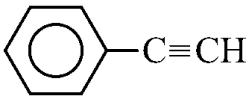
(A) NH_3 with CuO

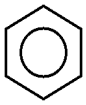
(B) NH_4NO_3

(C) $(NH_4)_2Cr_2O_7$

(D) $Ba(N_3)_2$

14) Which of the followign shows the positive bromine water test but not tollen's test :

(A) 

(B) 

(C) CH_3–CH=CH_2

(D) CH_3–CH_2–C≡CH

15) In the Reaction, $CH_3COOH \xrightarrow{LiAlH_4} A \xrightarrow{PCl_5} B \xrightarrow{Alc.KOH} C$, the product C is:

(A) Acetaldehyde

(B) Acetylene

(C) Ethylene

(D) Acetyl chloride

16) Reagent used to convert allyl alcohol to acrolein is :-

(A) MnO_2

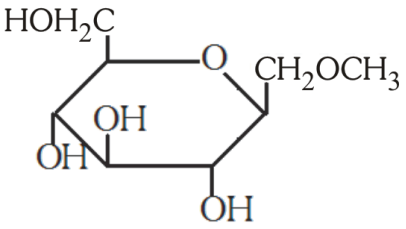
(B) $KMnO_4$

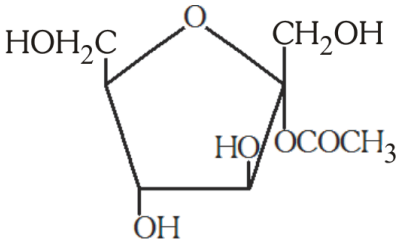
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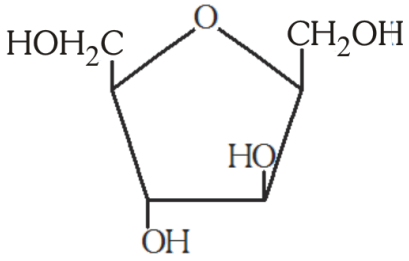
(C) OsO_4

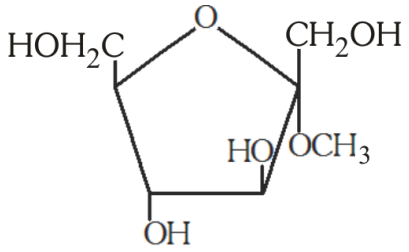
(D) H_2O_2

17) Which of the following compounds will behave as a reducing sugar in an aqueous KOH solution?

(A) 

(B) 

(C) 

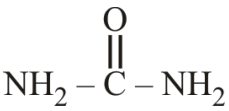
(D) 

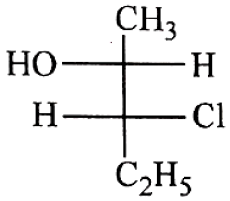
18) Kjeldahl's method cannot be used to estimate nitrogen for which of the following compounds?

(A) $C_6H_5NO_2$

(B) $C_6H_5NH_2$

(C) CH_3CH_2–C≡N

(D) 

19) IUPAC name of the compound 

(A) (2R,3R)-3-chloropentan-2-ol

(B) (2R,3S)-3-chloropentan-2-ol

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(C) (2S,3R)-3-chloropentan-2-ol

(D) (2S,3S)-3-chloropentan-2-ol

20) **Assertion :** Phenol gives phenopthalein after reaction with isopthalic acid
Reason : Phenopthalein is use as indicator.

(A) If both assertion and reason are true and the reason is a correct explanation of assertion.

(B) If both assertion and reason are true but reason is not a correct explanation of assertion.

(C) If assertion is true but the reason is false.

(D) If assertion is false but the reason is true.

SECTION-II

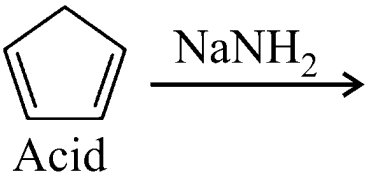
1) 0.5 cm thick coating (density = 1.2 g/ml) of silver is deposited on a plate of 0.06 m^2 area. The number of silver atoms deposited on plate are _____ $\times 10^{24}$.
(Given : Atomic mass of silver = 108, Avogadro's number = 6×10^{23})

2) The conductivity of saturated solution of Ag_3PO_4 is 3.06×10^{-6} Scm^{-1} and its molar conductance is 1.53 Scm^2 $mole^{-1}$. The K_{sp} of Ag_3PO_4 will be 4.32×10^{-x} M^4. Find value of x.

3) How many of the following sets are having correct order?
(i) $OCl_2 > SCl_2$ (Bond angle)
(ii) $NCl_3 > NF_3$ (Bond angle)
(iii) $OCl_2 > O(SiH_3)_2$ (Bond angle at oxygen)
(iv) $O(CH_3)_2 > O(SiH_3)_2$ (Bond angle)
(v) $CHCl_3 > CHF_3$ (Acidic Strength)
(vi) $BCl_3 > BBr_3$ (Lewis Acid Strength)

4) $NaBH_4 + I_2 \rightarrow$ Products
How many total moles of gaseous products are obtained using 6 moles of $NaBH_4$ and 3 moles I_2

5)



Conjugate base

How many contributing resonating structures of monoanion conjugate base are possible.

PART-3 : MATHEMATICS

SECTION-I

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1) Minimum value of $\frac{1}{2}\left(\cot\frac{A}{2}+3\tan\frac{A}{2}\right)$ where $A \in (0, 180°)$ occurs when A equals :-

(A) 60°

(B) 90°

(C) 120°

(D) 150°

2) Let F_1, F_2 are the foci of the hyperbola $\frac{x^2}{16}-\frac{y^2}{9}=1$ and F_3, F_4 are the foci of its conjugate hyperbola. If e_H and e_C are their eccentricities respectively then the statement which holds true is :-

(A) Their equations of the asymptotes are different.

(B) $e_H > e_C$

(C) Area of the quadrilateral formed by their foci is 50 sq. units.

(D) Their auxiliary circles will have the same equation.

3) **Assertion (A) :** A light ray moving along the straight line y = 5 after reflecting from the concave mirror $y^2 - 8y - x + 19 = 0$ passes through the point $\left(\frac{13}{4}, 4\right)$.
and
Reason (R) : Any light ray moving parallel to axis of a parabola after reflecting from parabola passes through the focus of the parabola.

(A) **A** is correct but **R** is not correct.

(B) **A** is not correct but **R** is correct.

(C) Both **A** and **R** are correct but **R** is **NOT** the correct explanation of **A.**

(D) Both **A** and **R** are correct and **R** is the correct explanation of **A.**

4) Let $\Delta_0 = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ and let Δ_1 denote the determinant formed by the cofactors of elements of Δ_0 and Δ_2 denote the determinant formed by the cofactor at Δ_1 and so on Δ_n denotes the determinant formed by the cofactors at Δ_{n-1} then the determinant value of Δ_n is :-

(A) Δ_0^{2n}

(B) $\Delta_0^{2^n}$

(C) $\Delta_0^{n^2}$

(D) Δ_0^2

5) If $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ and $\det(A^n - I) = 1 - \lambda^n$, $n \in N$ then the value of λ, is :-

(A) 1

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(B) 2

(C) 3

(D) 4

6) For each point (x, y) on the ellipse with centre at the origin and principal axes along the coordinate axes, the sum of the distances from the point (x, y) to the points (± 2, 0) is 8. The positive value of x such that (x, 3) lies on the ellipse, is :-

(A) $\frac{\sqrt{3}}{3}$

(B) 2

(C) 4

(D) $2\sqrt{3}$

7) A line makes an intercept of 5 and 4 on the x-axis & y-axis respectively. If the axes are rotated about the origin by an angle α such that the length of intercept on the new x-axis is 8. The length of intercept on the new y-axis, is

(A) $\sqrt{23}$

(B) $\frac{20}{8}$

(C) $\frac{40}{\sqrt{139}}$

(D) 1

8) In the ΔABC, medians AD and BE are drawn. If AD = 4, ∠BAD = π/6 and ∠ABE = π/3, then area of the ΔABC is

(A) 64/3

(B) 8/3

(C) 16/3

(D) $32/3\sqrt{3}$

9)

List-I		List-II	
(A)	If $a^2 + b^2 + c^2 = 0$ and martix $A = \begin{bmatrix} b^2+c^2 & ab & ac \\ ab & c^2+a^2 & bc \\ ac & bc & a^2+b^2 \end{bmatrix}$ and if $\lvert adj(adjA)\rvert = 32\lambda a^8b^8c^8$, (a, b, c ≠ 0), then λ is equal to	(I)	1
(B)	If x is positive and matrix $\begin{bmatrix} x & 1 \\ 2 & 2x \end{bmatrix}$ is invertible, then x cannot be equal to	(II)	0

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(C)	The value of m so that matrix $\begin{bmatrix} {}^{10}C_4 & {}^{10}C_5 & {}^{11}C_m \\ {}^{11}C_6 & {}^{11}C_7 & {}^{12}C_{m+2} \\ {}^{12}C_8 & {}^{12}C_9 & {}^{13}C_{m+4} \end{bmatrix}$ is singular, is	(III)	8
(D)	If $a\lambda^4 + b\lambda^3 + c\lambda + d\lambda + e = \begin{vmatrix} \lambda^2+3\lambda & \lambda-1 & \lambda+3 \\ \lambda+1 & 2-\lambda & \lambda-3 \\ \lambda & 3 & \lambda+4 \end{vmatrix} - 21$ then e is equal to	(IV)	5

(A) (A) - (III); (B) - (I,II); (C) - (IV); (D) - (II)

(B) (A) - (I); (B) - (I,II); (C) - (II); (D) - (IV)

(C) (A) - (II); (B) - (I); (C) - (III); (D) - (IV)

(D) (A) - (IV); (B) - (II); (C) - (I); (D) - (III)

10) The number of solutions of the equation $\sqrt{2}\cos 3x - \cos 2x = \sqrt{3}\sin 2x + \sqrt{2}\sin 3x$ in $\left[\frac{-4\pi}{3}, \frac{2\pi}{3}\right]$ is

(A) 3

(B) 4

(C) 5

(D) 6

11) The slope of the three sides of a triangle PQR are –1, –2 and 3. If the orthocentre of the triangle is at origin, then the locus of the centroid is given by :

(A) $9x = 2y$

(B) $9x = -2y$

(C) $2x = 9y$

(D) $4x = 9y$

12) The length of the shortest chord of the circle $x^2 + y^2 - 4x + 6y - 3 = 0$ which passes through the point P(1, –1) is :

(A) $\sqrt{11}$

(B) $2\sqrt{11}$

(C) $\sqrt{10}$

(D) $\sqrt{5}$

13) Consider a circle S : $(x + 2)^2 + y^2 = 1$ and a parabola P : $y^2 = 4x$, if n is number of common tangents to both the curve and $m_1, m_2 \ldots m_n$ are the slopes of these tangents, then $\sum_{i=1}^{n} m_i^2$ equals :

(A) $\frac{5}{3}$

(B) $\frac{10}{3}$

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(C) $\frac{20}{3}$

(D) 10

14) A line of fixed length 2 units moves so that its one end is on the positive x-axis and other end on that part of the line x + y = 0, which lies in second quadrant. The locus of the mid-point of the line is given by

(A) $x^2 + 5y^2 + 4xy - 1 = 0$

(B) $x^2 + 5y^2 + 4xy + 1 = 0$

(C) $x^2 + 5y^2 - 4xy - 1 = 0$

(D) $4x^2 + 5y^2 + 4xy + 1 = 0$

15) **Statement-I :** If $3x + 2y + z \neq 0$ & $(2x + y)^3 + (x + 2z)^3 + (y - z)^3 - 3(2x + y)(x + 2z)(y - z) = 0$, then $2x + y = x + 2z = y - z$
and
Statement-II : $a^3 + b^3 + c^3 - 3abc = 0 \Rightarrow a = b = c$

(A) Statement-I is true but Statement-II is false.

(B) Both Statement-I and Statement-II are false.

(C) Both Statement-I and Statement-II are true.

(D) Statement-I is false but Statement-II is true.

16) The lines 5x + 12y - 10 = 0 and 5x - 12y - 40 = 0 touch a circle of radius 3 units. If the centre of circle lies in the first quadrant then the coordinates of centre is

(A) (5, 3)

(B) (5, 1)

(C) (5, 2)

(D) (5, 6)

17) If the minimum distance between the parabola $y^2 - 4x - 8y + 40 = 0$ and $x^2 - 8x - 4y + 40 = 0$ is d. find d^2.

(A) 1

(B) 0

(C) 3

(D) 2

18) Variable ellipses are drawn with x = -4 as a directrix and origin as corresponding focus. The locus of extremities of minor axes of these ellipses is

(A) $y^2 = 4x$

(B) $y^2 = 2x$

(C) $y^2 = x$

(D) $x^2 = 4y$

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19) If $x\cos\alpha + y\sin\alpha = P$, be a variable chord of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{2a^2} = 1$ subtends a right angle at the centre of the hyperbola, then the chords touch a fixed circle with centre origin and radius r, then r =

(A) a

(B) $\sqrt{2}a$

(C) 2a

(D) 4a

20) Given a quadratic equation $x^2 - (a + b)x + ab = 0$, now consider a square matrix A of order 3 × 3 such that :

1	Each element a_{ii} = sum of roots of given equations
2	Each element $a_{i,\ i+1}$ = product of roots
3	Each element $a_{i,\ i-1}$ = unity
4	Rest all elements = zero

then |Adj A| = _____

(A) $(a^2 + b^2)^2$

(B) $(a - b)^2$

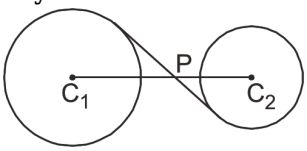
(C) $(a^2 + b^2)^2 (a - b)^2$

(D) $(a^2 + b^2)^2 (a + b)^2$

SECTION-II

1) Number of integral values of x the inequality $\log_{10}\left(\frac{2x-2007}{x+1}\right) \le 0$ holds true, is

2) In the figure given, two circles with centres C_1 and C_2 are 35 units apart, i.e. $C_1C_2 = 35$. The radii of the circles with centres C_1 and C_2 are 12 and 9 respectively. If P is the intersection of C_1C_2 and a common internal tangent to the circles, then C_1P equals :



3) For what value of λ, the system of equations x + y + z = 6, x + 2y + 3z = 10, 2x + 4y + λz = 24 is inconsistent :

4) Let A and B be two square matrices of same order such that $A^3 + B^3 = 0$ and $A^2B + AB^2 = 0$ but $|A| \ne |B|$. Then $|A^2 - B^2|$ is equal to

5) If length of perpendiculars drawn from focii of an ellipse E to any of its tangents are p_1 and p_2. If ellipse has its centre at C(1, 3) and focus at S(6, 3). also P(4, 7) lies on the ellipse then find $p_1\ p_2$ = __

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ANSWER KEYS

PART-1 : PHYSICS

SECTION-I

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	A	C	A	A	A	C	B	D	B	C	A	B	D	A	D	C	D	A	A	C

SECTION-II

Q.	21	22	23	24	25
A.	3	625	6	3	12

PART-2 : CHEMISTRY

SECTION-I

Q.	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
A.	C	C	A	A	D	A	A	B	A	D	C	C	D	C	C	A	B	A	A	D

SECTION-II

Q.	46	47	48	49	50
A.	2	10	3	6	5

PART-3 : MATHEMATICS

SECTION-I

Q.	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70
A.	A	C	D	B	B	B	C	D	A	D	C	B	B	A	A	C	D	A	B	D

SECTION-II

Q.	71	72	73	74	75
A.	1005	20	6	0	20

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SOLUTIONS

PART-1 : PHYSICS

1)

Mass number of daughter nucleus (M) = 208 – 4 = 204
Now, total energy of disintegration = energy of daughter nucleus + energy of α-particle

or $E_t = \frac{p^2}{2m} + \frac{P^2}{2m}$

Since momentum is conserved, p = P. Hence

$E_r = \frac{p^2}{2}\left(\frac{1}{m} + \frac{1}{M}\right) = \frac{p^2}{2}\left(\frac{m+M}{mM}\right)$

Energy of α-particle $= \frac{p^2}{2m} = E$ Hence

$E_t = \left(\frac{m+M}{M}\right)E$

$= \left(\frac{4+204}{204}\right)E = \frac{52}{51}E$

2) $K = \frac{1240}{620} - 1.5 = 0.5\ eV = eEd$

$= e \times \frac{2\epsilon_0}{\epsilon_0} d$

$d = \frac{1}{4}m$

3) For same mass surface area in maximum for thin circular plate and rest parameter depends on material.

4) $\frac{dN}{dt} \times \frac{hC}{\lambda} = P$

$F = \frac{dN}{dt}\frac{h}{\lambda} = ma$

$S = \frac{1}{2}at^2$

$S = \frac{f}{2} \times \frac{P}{mc} \times t^2$

$= \frac{1}{2} \times \frac{3 \times 10^{-3}}{50 \times 10^{-3} \times 3 \times 10^8} \times (3600)^2$

$= \frac{36 \times 36 \times 10^4}{10^{10}} \times 10^{-6} = 1.3mm$

5) Due to a disc, electric field on axis is given by

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$$E = \frac{\sigma}{2\epsilon_0}\left[1 - \frac{x}{\sqrt{R^2 + x^2}}\right]$$

$$\therefore E = \frac{\sigma}{2\epsilon_0}\left[1 - \frac{h}{\sqrt{(2a)^2 + h^2}}\right] - \frac{\sigma}{2\epsilon_0}\left[1 - \frac{h}{\sqrt{a^2 + h^2}}\right]$$

$$= \frac{\sigma}{2\epsilon_0}\left[\frac{h}{2a\sqrt{1 + \frac{h^2}{4a^2}}} - \frac{h}{a\sqrt{1 + \frac{h^2}{a^2}}}\right]$$

$$= \frac{\sigma}{2\epsilon_0}\frac{h}{a}\left[\frac{1}{2}\left(1 - \frac{h^2}{2 \times 4a^2}\right) - \left(1 - \frac{1h^2}{2a^2}\right)\right]$$

$$= \frac{ah}{2\epsilon_0 a}\left[\frac{1}{2} - 1\right] = \frac{\sigma h}{4\epsilon_0 a}$$

6) $\phi = \vec{E} \cdot \vec{A}$

$(E_0\hat{i} + E_0\hat{j}) \cdot (-a^2 \cos 30^\circ \hat{k} + a^2 \sin 30^\circ \hat{i})$

$\frac{E_0 a^2}{2}$

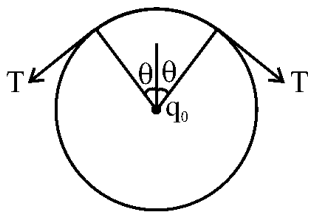
7) $E_x \times 4\pi x^2 = \frac{\rho_0 \int_0^x x^2 \times 4\pi x^2 dx}{\epsilon_0} = \frac{4\pi\rho_0 x^5}{5\epsilon_0}$

$E_y = \frac{\rho_0 \int_0^R x^2 \times 4\pi x^2 dx}{\epsilon_0 \times 4\pi y^2} = \frac{\rho_0 \times R^5}{5\epsilon_0 y^2}$

$\frac{\rho_0 x^3}{5\epsilon_0} = \frac{\rho_0 R^5}{5\epsilon_0 y^2}$

$x^3y^2 = R^5$

8)



θ is small, then

$2T \sin\theta = kq^0\left(\frac{Q}{2\pi R}\right)\frac{(R2\theta)}{R^2}$

Putting $\sin\theta = \theta$ we get

$T = \frac{1}{8\pi^2\epsilon_0}\frac{Qq_0}{R^2}$

Now $Y = \frac{TR}{A\Delta R}$

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$\therefore\ \Delta R = \frac{TR}{AY} = \left(\frac{1}{8\pi^2\epsilon_0}\frac{Qq_0}{R}\right)\left(\frac{1}{AY}\right)$

9)

For (A) : $W = \frac{1}{2}(2P_0)(V_0) = P_0V_0$

For (B) : $Q_{AB} = nC_v(T_B - T_A) = 1 \times \frac{3}{2}R\left(\frac{3P_0V_0 - P_0V_0}{R}\right) = 3P^0V^0$

For (C) : $T_A = \frac{P_0V_0}{R}$, $T_B = \frac{3P_0V_0}{R} = 3T_A$, $T_C = \frac{P_0(2V_0)}{R} = 2T_A$

$\Rightarrow$ Minimum temperature : $T_{min} = \frac{P_0V_0}{R}$

(OR Temperature will be minimum for a point which is nearest to origin)

Maximum temperature will be either on end points or in process BC.

For (D) : As for process BC : $\frac{P-3P_0}{V-V_0} = \frac{2P_0}{-V_0} \Rightarrow P = 5P^0 - \left(\frac{2P_0}{V_0}\right)V$

$\Rightarrow PV = 5P^0V - \left(\frac{2P_0}{V_0}\right)V^2$ But $PV = RT$. So $RT = 5P^0V - \left(\frac{2P_0}{V_0}\right)V^2$

$\Rightarrow R\left(\frac{dT}{dV}\right) = 5P^0 - \left(\frac{2P_0}{V_0}\right)(2V) = 0 \Rightarrow V = \frac{5}{4}V_0$

$\Rightarrow RT_{max} = 5P^0\left(\frac{5}{4}V_0\right) - \left(\frac{2P_0}{V_0}\right)\left(\frac{5}{4}V_0\right)^2 \Rightarrow T_{max} = \frac{25P_0V_0}{8R}$

10) $\left.\frac{dm}{dt}\right|_{Steam} = \frac{1}{16}\left.\frac{dm}{dt}\right|_{ice}$

$\left.\frac{dQ}{dt}\right|_{Steam} = \frac{T-100}{\left(\frac{2\ell}{3Ak}\right)} = \left(\frac{dm}{dt}\right)_{steam}L_v$

$\left.\frac{dQ}{dt}\right|_{Ice} = \frac{T-0}{\left(\frac{\ell}{3Ak}\right)} = 16\left(\frac{dm}{dt}\right)_{steam}L_f$

$\frac{\frac{T-100}{2}}{\frac{T}{T}} = \frac{L_v}{16L_f}$

$\frac{(T-100)}{2T} = \frac{540}{16 \times 80}$

$\frac{T-100}{T} = \frac{540}{640}$

$\frac{T-100}{2T} = \frac{540}{8 \times 80}$

11) $\Delta S = nC\ln\frac{T_b}{T} + \frac{mL}{T_b} = 697\ J/K$

12)

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At $V = V_0$

$$P = \frac{P_0}{1 + \left(\frac{V_0}{V_0}\right)^3} = \frac{P_0}{2}$$

$$T.K.E. = \frac{3}{2}PV = \frac{3}{2}\frac{P_0}{2} \times V_0$$

$$= \frac{3}{4}P_0V_0$$

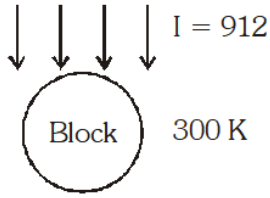
13) Amount of energy absorbed by H_2 gas = 10 × 80 = 800 cal

Degree of freedom of H_2 gas is 5.

Translational KE of H_2 gas is = $\frac{3}{5} \times 800$ = 480 cal

14) Ut ad study state T

$\sigma 4\pi R^2 T^4 = \sigma 4\pi R^2 T_0^4 + I\pi R^2$



$$T^4 = T_0^4 + \frac{I}{4\sigma}$$

$$= (300)^4 + \frac{912}{4 \times 5.7 \times 10^{-8}}$$

$= (300)4 + 40 \times 108$

$T^4 = 10^8 (121)$

$T = 100 \times 3.31$

$= 331$

$T = 330$

15) $\phi = 0 \Rightarrow q_{net} = 0 \Rightarrow \int \vec{E}.d\vec{s} = 0$. This does not necessarily implies that E = 0.

16)

Truth table for NAND gate

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

17) diode acts is short circuit

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$\therefore i = \frac{2}{80}A$

18) $\theta = \omega_1 t$

$\theta + \pi = \omega_2 t$

$\pi = (\omega_2 - \omega_1)t$

$t = \frac{\pi}{\omega_2 - \omega_1}$

& $\omega = \sqrt{\frac{GM}{r^3}}$

19) Orbital speed $v_0 = \sqrt{\frac{Gm}{R}}$

$2 = \sqrt{\frac{Gm}{R}}$

Hence escape speed $v_e = \sqrt{\frac{2Gm}{R}} = 2\sqrt{2}$ m/s

20) $T_1\cos30° + T_2\cos60° = mg$; $T_1\sin30° = T_2\sin60°$

Solving these equations we get $T_1 = \frac{\sqrt{3}}{2}mg$, $T_2 = \frac{mg}{2}$;

$\frac{\text{stress in wire - 1}}{\text{stress in wire - 2}} = \frac{T_1A_2}{T_2A_1} = \frac{\frac{\sqrt{3}}{2}mg \times \sqrt{3}}{\frac{mg}{2} \times 3}$

21) $\Delta E_{2-1} = 13.6 \times Z^2\left[1 - \frac{1}{4}\right] = 13.6 \times Z^2\left[\frac{3}{4}\right]$

$\Delta E_{3-2} = 13.6 \times Z^2\left[\frac{1}{4} - \frac{1}{9}\right]$

$= 13.6 \times Z^2\left[\frac{5}{36}\right]$

$\therefore \Delta E_2 = \Delta E_{3-2} + 74.8$

$13.6 \times Z^2\left[\frac{3}{4}\right] = 13.6 \times Z^2\left[\frac{5}{36}\right] + 74.8$

$13.6Z^2\left[\frac{3}{4} - \frac{5}{36}\right] = 74.8$

$Z^2 = 9$

$\therefore Z = 3$

22) Energy of incident photons

$\frac{6.62 \times 10^{-34} \times 1.6 \times 10^{15}}{1.6 \times 10^{-19}}$eV $= 6.62$ eV

No. of photons $= \frac{6.62}{1.6 \times 10^{-19} \times 6.62}$

$= 6.25 \times 10^{18}$

$K = 6.25$

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23)

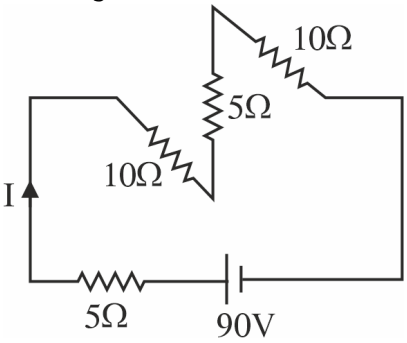
$T_1 = 300K$, $T_2 = 277K$

$$\frac{Q_2}{1} = \frac{T_2}{T_1 - T_2}$$

$$Q2 = \frac{250}{300 - 250} = 5J$$

$Q_1 = W + Q_2 = 1 + 5 = 6J$

24) In given circuit both the diodes will be reverse biased so effective circuit is shown below



The current in circuit is given as

$$I = \frac{90}{(5 + 10 + 5 + 10)} = \frac{90}{30}A$$

25) $\frac{\Delta v}{v} = \gamma \Delta T$

$$\beta = \frac{\Delta P}{\frac{\Delta v}{v}} \Rightarrow \Delta P = \beta(\gamma \Delta T)$$

$= 10^4 \times 3 \times 2 \times 10^{-5} \times 20 = 12$ Pa

PART-2 : CHEMISTRY

26) Number of days $= \frac{\frac{8 \times 24.8}{248} \times 6 \times 10^{23}}{60 \times 10^6} = 8 \times 10^{15}$

27) Higher van't Hoff factor lesser freezing point for same concentration solution.

28) $\Delta S = 1 \times \frac{5R}{2} \ln 2 + 1 \times R \ln \frac{1}{4} = \frac{R \ln 2}{2}$

30) $\Delta T_b = K_b \times m$

$$m = \frac{102 - 100}{0.5} = 4$$

$$\frac{P^o - P^s}{P^s} = \frac{m \times M}{1000}$$

$$\frac{760 - P^s}{P^s} = \frac{4 \times 18}{1000}$$

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P^s = 708.955 mmHg.

32) Conceptual

35) Conceptual

37) Bead produced with CaSOa is colourless $Ca(BO_2)_2$, because Ca^{2+} is a colourless ion.

38) $Ba(N_3)_2 \xrightarrow{Heat} Ba(s) + 3N_2(g)$

Azide salt of barium can be obtained in purest form as well as the decomposition product contain solid Ba as by product
along with gaseous nitrogen, hence no additional step of separation is required.
Other reactions are

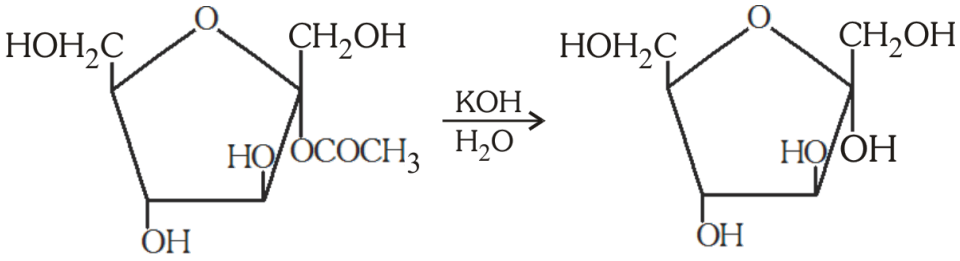
$$NH_4NO_3 \xrightarrow{Heat} N_2O + 2H_2O$$

$$2NH_3 + 3CuO \xrightarrow{Heat} 3Cu + 3H_2O + N_2$$

$$(NH_4)_2Cr_2O_7 \xrightarrow{Heat} Cr_2O_3 + 4H_2O + N_2$$

40) $CH_3COOH \xrightarrow{LiAlH_4} CH_3 - CH_2 - OH \xrightarrow{PCl_5}$
$CH_3CH_2 - Cl \xrightarrow[Alc\ KOH]{\Delta} CH_2 = CH_2$

41) Fact based.

42) 

43) Kjeldahl's method for estimation of nitrogen is not applicable for nitrobenzene $C_6H_5NO_2$. because reaction with H_2SO_4, nitrobenzene can not give ammonia.

46) Mass of silver deposited = $0.5 \times 0.06 \times 10^4 \times 1.2 = 360$ g

Number of silver atoms deposited = $\frac{360}{108} \times 6 \times 10^{23} = 2 \times 10^{24}$

47) $\lambda_m = \frac{\kappa \times 1000}{M}$

$1.53 = \frac{3.06 \times 10^{-6} \times 1000}{M}$

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$M = 2 \times 10^{-3}$

$Ag_3PO_4(s) \rightleftharpoons 3Ag^+(aq) + PO_4^{3-}(aq)$

$\quad\quad\quad\quad\quad\quad 3M \quad\quad\quad M$

$K_{SP} = 27M^4 = 27 \times (2 \times 10^{-3})^4 = 4.32 \times 10^{-10}\ M^4$

48) (i), (ii), (v)

49) $2NaBH_4 + I_2 \rightarrow B_2H_6(g) + 2NaI + H_2(g)$

$\Rightarrow$ Total 6 moles of gaseous products

PART-3 : MATHEMATICS

51) Using AM $\geq$ GM

$$\frac{1}{2}\left(\cot\frac{A}{2} + 3\tan\frac{A}{2}\right) \geq \sqrt{3}$$

Hence minimum value is $\sqrt{3}$

now, $\cot\frac{A}{2} + 3\tan\frac{A}{2} = 2\sqrt{3}$; $3\tan^2\frac{A}{2} - 2\sqrt{3}\tan\frac{A}{2} + 1 = 0$

$$\tan\frac{A}{2} = \frac{2\sqrt{3} \pm \sqrt{12-12}}{6} = \frac{1}{\sqrt{3}};\ \frac{A}{2} = 30°$$

$\Rightarrow A = 60°$

52) $e_H = 5/4$; $e_C = 5/3$ area $= \frac{d_1 d_2}{2} = \frac{100}{2} = 50$

AC : $x^2 + y^2 = 16$; AH = $x^2 + y^2 = 9$

53)

Equation of parabola can be written as $(y - 4)^2 = (x - 3)$

and its equation of axis is $y = 4$ & its focus $\left(\frac{13}{4}, 4\right)$

hence both Statements are true.

54) $\Delta_1 = \Delta_0^2$ $\Delta_2 = \Delta_1^2 = \Delta_0^4$

$\Delta_3 = \Delta_2^2 = \Delta_0^8$

$\Delta_4 = \Delta_3^2 = \Delta_0^{16}$

and so on $\Delta_n = \Delta_0^{2^n}$

55) $A^n - I = \begin{bmatrix} 2^{n-1} - 1 & 2^{n-1} \\ 2^{n-1} & 2^{n-1} - 1 \end{bmatrix}$

hence $|A^n - I| = (2^{n-1} - 1)^2 - (2^{n-1})^2$

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$= (2^{n-1} - 1 - 2^{n-1})(2^{n-1} - 1 + 2^{n-1})$

$= 1 - 2^n$

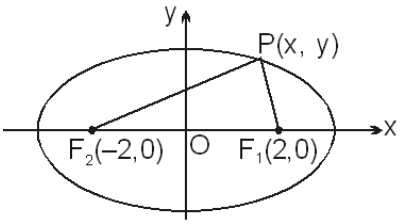
Hence, $\lambda = 2$

56) $c = 2; \quad a = 4$

equation of the ellipse is $\frac{x^2}{a^2} + \frac{y^2}{a^2 - a^2e^2} = 1$

$\frac{x^2}{16} + \frac{y^2}{16-4} = 1$

$\Rightarrow \frac{x^2}{16} + \frac{y^2}{12} = 1$



when y = 3 then $\frac{x^2}{16} + \frac{9}{12} = 1$

$\Rightarrow \frac{x^2}{16} = \frac{1}{4}$

$\Rightarrow x = 2$

57) Distance from origin remains same

$\frac{1}{4^2} + \frac{1}{5^2} = \frac{1}{8^2} + \frac{1}{k^2}$

$\Rightarrow k = \pm\frac{40}{\sqrt{139}}$

$|k| = \frac{40}{\sqrt{139}}$

58) AD = 4

$AP = \frac{2}{3} \times 4 = \frac{8}{3}$

In ΔABP

$\frac{BP}{AP} = \tan\frac{\pi}{6} \Rightarrow BP = \frac{8}{3\sqrt{3}}$

Area of ΔABC = 3 × Area of ΔABP = $\frac{32}{3\sqrt{3}}$

59)

(A) If $a^2 + b^2 + c^2 = 0$ and

$$A = \begin{bmatrix} b^2 + c^2 & ab & ac \\ ab & c^2 + a^2 & bc \\ ac & bc & a^2 + b^2 \end{bmatrix}$$

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$$|A| = \begin{vmatrix} -a^2 & ab & ac \\ ab & -b^2 & bc \\ ac & bc & -c^2 \end{vmatrix} = a^2b^2c^2 \begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix}$$

$|A| = 4a^2b^2c^2$

$|\text{adj}(\text{adj}A)| = |A|^{(n-1)^2} = |A|^4 = 256a^8b^8c^8$

$256\, a^8b^8c^8 = 32\lambda a^8b^8c^8$

$\lambda = 8$

(B) $x > 0$ and $\begin{bmatrix} x & 1 \\ 2 & 2x \end{bmatrix}$ is invertible

$\Rightarrow 2x^2 - 2 \neq 0 \Rightarrow x \neq \pm 1$

hence x can not take the values 0 & 1.

(C) $\begin{bmatrix} {}^{10}C_4 & {}^{10}C_5 & {}^{11}C_m \\ {}^{11}C_6 & {}^{11}C_7 & {}^{12}C_{m+2} \\ {}^{12}C_8 & {}^{12}C_9 & {}^{13}C_{m+4} \end{bmatrix}$

$C_1 + C_2 \to C_1$

$\begin{bmatrix} {}^{11}C_5 & {}^{10}C_5 & {}^{11}C_m \\ {}^{12}C_7 & {}^{11}C_7 & {}^{12}C_{m+2} \\ {}^{13}C_9 & {}^{12}C_9 & {}^{13}C_{m+4} \end{bmatrix}$

the value of m is 5

(D) To find e put $\lambda = 0$

$$e = \begin{vmatrix} 0 & -1 & 3 \\ 1 & 2 & -3 \\ -3 & 4 & 0 \end{vmatrix} - 21$$

$e = 0$

60) $\sqrt{2}(\cos 3x - \sin 3x) = \sqrt{3}\sin 2x + \cos 2x$

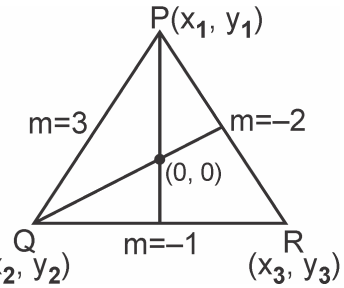
$$\frac{(\cos 3x - \sin 3x)}{\sqrt{2}} = \left(\frac{\sqrt{3}\sin 2x + \cos 2x}{2}\right)$$

$$\cos\left(3x + \frac{\pi}{4}\right) = \cos\left(2x - \frac{\pi}{3}\right)$$

$$3x + \frac{\pi}{4} = 2n\pi \pm \left(2x - \frac{\pi}{3}\right)$$

$$x = 2n\pi - \frac{7\pi}{12}, x = \frac{2n\pi}{5} + \frac{\pi}{60}; n \in Z$$

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61) $\frac{y_1}{x_1} = 1, \quad \frac{y_2}{x_2} = \frac{1}{2}$

$\frac{y_3}{x_3} = -\frac{1}{3}$

Slopes of QR, RP and PQ are –1, –2, 3.

$m_{QR} = \frac{\frac{x_2}{2} + \frac{x_3}{3}}{x_2 - x_3} = -1 \quad \Rightarrow \quad 9x_2 = 4x_3$

Similarly $9x_1 = 5x_3$

$G = (h, k) = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{x_1 + \frac{x_2}{2} - \frac{x_3}{3}}{3} \right)$

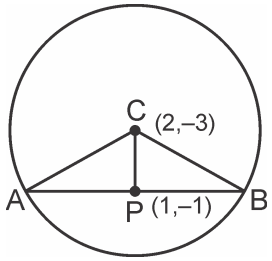
$\Rightarrow \quad \frac{k}{h} = \frac{2}{9}$

$9y = 2x$

62) $S = x^2 + y^2 - 4x + 6y - 3 = 0$

C(2, –3)

$r = \sqrt{4 + 9 + x} = 4$



Length of chord is shortest when CP is ⊥ to chord.

$AB = 2\sqrt{r^2 - cp^2}$

$= 2\sqrt{16 - 5} = 2\sqrt{11}$

63) Equation of tangent to the parabola $y = 4x$

$y = mx + \frac{1}{m}$ which is also tangent to the circle p = r.

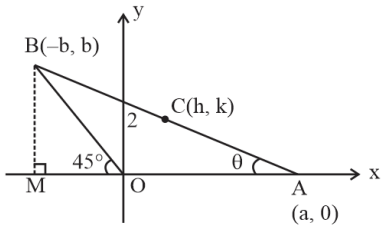
$\frac{\left| -2m + \frac{1}{m} \right|}{\sqrt{1 + m^2}} = 1$

$|1 - 2m^2| = m\sqrt{1 + m^2}$

$3m^4 - 5m^2 + 1 = 0$

$\sum_{i=1}^{4} m_i^2 = 4 \times \frac{5}{6} = \frac{10}{3}$

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64)

$\Rightarrow BM = 2\sin\theta,\ MA = 2\cos\theta$

$\Rightarrow OM = 2\sin\theta \Rightarrow B \equiv (-2\sin\theta,\ 2\sin\theta)$

$\Rightarrow A \equiv (2\cos\theta - 2\sin\theta,\ 0)$

Now, $h = \dfrac{2\cos\theta - 2\sin\theta - 2\sin\theta}{2}$

$\Rightarrow \cos\theta - 2\sin\theta = h$ ..... (i)

$k = \dfrac{0 + 2\sin\theta}{2} \Rightarrow \sin\theta = k$ .... (ii)

From (i)

$\Rightarrow \cos\theta = h + 2k$ ..... (iii)

$(ii)^2 + (iii)^2$

$\Rightarrow 1 = k^2 + (h + 2k)^2$

$\Rightarrow x^2 + 5y^2 + 4xy = 1$

65) I is true, II is false.

$a^3 + b^3 + c^3 - 3abc$

$= (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$

$= \dfrac{1}{2}(a+b+c)[(a-b)^2 + (b-c)^2 + (c-a)^2]$

Hence, $a^3 + b^3 + c^3 - 3abc = 0$

⇒ Either $a + b + c = 0$ or $a = b = c$

with the help of this result we can easily verify the statements.

66) Let, centre $\equiv (h,\ k)$

$\left|\dfrac{5h + 12k - 10}{13}\right| = 3$ and $\left|\dfrac{5h - 12k - 40}{13}\right| = 3$

$\Rightarrow 5h + 12k - 10 = \pm 39$ ..... (i)

$\Rightarrow 5h - 12k - 40 = \pm 39$ ..... (ii)

Solving (i) & (ii)

$\Rightarrow h = 5,\ k = 2$

67) Both parabola are symm. about line y = x.

⇒ Minimum distance is between tangents to parabola parallel to y = x.

$\Rightarrow x^2 - 8x - 4y + 40 = 0$

⇒ differentiate w.r.t x.

$\Rightarrow 2x - 8 - 4y^1 = 0$

$\Rightarrow y^1 = \dfrac{x-4}{2} = 1 \Rightarrow x = 6$

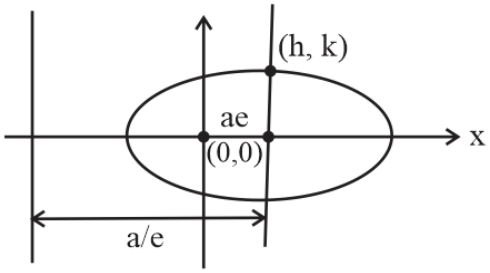
$\Rightarrow y = 7 \Rightarrow P \equiv (6,\ 7)$

Similarly, $Q \equiv (7,\ 6)$

$d = \sqrt{1^2 + 1^2} = \sqrt{2}$

$d^2 = 2.$

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68) $x = -4$

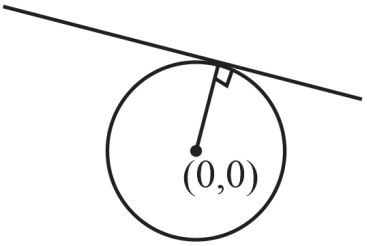
$\Rightarrow k = b,\ h = ae$ and $a/e - ae = 4$

$\Rightarrow a\left(1-e^2\right) = 4e \Rightarrow a^2\left(1-e^2\right) = 4ae$

$\Rightarrow b^2 = 4h$

$\Rightarrow k^2 = 4h \Rightarrow y^2 = 4x$

69) Using concept of homogenising,



$$\frac{x^2}{a^2} - \frac{y^2}{2a^2} - 1^2 = 0$$

$$\Rightarrow \frac{x^2}{a^2} - \frac{y^2}{2a^2} - \left(\frac{x\cos\alpha + y\sin\alpha}{P}\right)^2 = 0$$

$\theta = 90^\circ$

$\Rightarrow$ coefficient of x^2 + coefficient of $y^2 = 0$

$$\Rightarrow \frac{1}{a^2} - \frac{\cos^2\alpha}{P^2} - \frac{1}{2a^2} - \frac{\sin^2\alpha}{P^2} = 0$$

$$\Rightarrow \frac{1}{2a^2} = \frac{1}{P^2} \Rightarrow P = \sqrt{2}a$$

$\Rightarrow$ radius $= \sqrt{2}a$

70) $|A| = \begin{vmatrix} a+b & ab & 0 \\ 1 & a+b & ab \\ 0 & 1 & a+b \end{vmatrix} = (a^2 + b^2)(a + b)$

$|\text{Adj } A| = |A|^2 = ((a^2 + b^2)(a + b))^2$

71) $0 < \dfrac{2x-2007}{x+1} \le 1$

$\dfrac{2x-2007}{x+1} > 0$ and $\dfrac{2x-2007}{x+1} \leqslant 1$

$x \in (-\infty - 1) \cup (1003.5, \infty)$ ....(i)

$x \in (-1, 2008]$ ....(ii)

from (i) and (ii)

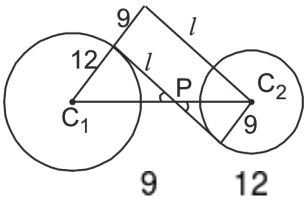
$x \in (1003.5, 2008]$

No. of integers 1005

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72) let $C_1P = a$

$PC_2 = 35 - a$



Hence, $\frac{9}{35-a} = \frac{12}{a}$

$\Rightarrow 3a = 4(35 - a)$

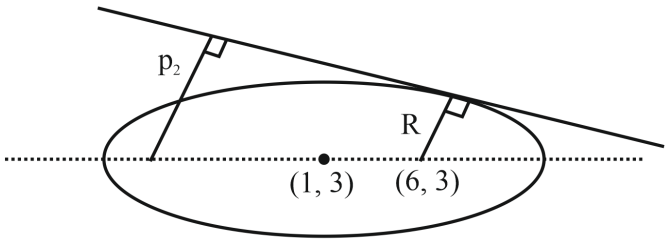
$\Rightarrow 7a = 4 \cdot 35$

$\Rightarrow a = 20$

73) Solve first two equations to get $x = 2 + z$ and $y = 4 - 2z$. Put in third equation to get $(\lambda - 6)z = 4$. No solution for $\lambda = 6$.

OR $\begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & 4 & \lambda \end{vmatrix} = 0 \Rightarrow \lambda = 6.$

74) $A^2B + AB^2 = 0$. Pre-multiply both sides by A to get $A^3B + A^2B^2 = (-B^3)B + A^2B^2 = (-B^2 + A^2)B^2 = 0$. Take determinant on both sides to get $|A^2 - B^2| = 0$ or $|B| = 0$. If $|B| = 0$ then taking determinant on both sides of $A^3 = -B^3$ gives $|A| = 0$, which is contradiction.



75)

$ae = 5$ .... (1)

$\frac{(x-1)^2}{a^2} + \frac{(y-3)^2}{b^2} = 1$

P(4, 7) lies on it

$\frac{9}{a^2} + \frac{16}{b^2} = 1$ .... (2) &

$e^2 = 1 - \frac{b^2}{a^2}$ .... (3)

on solving $b^2 = 20$

= product of length of $\perp$ is = $p_1 p_2$

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